

The Cell Painting Shortcut Audit — what problem we are working on and what we are trying to solve

A deep problem statement for the team (biology and ML), written 3 October 2026, seven days before the 10 October deadline.

0. How to read this document

This document answers four questions in order: **why does this problem exist** (§1–3), **what exactly are we trying to decide, scientifically and statistically** (§4–5), **what have we built and found so far** (§6–7), and **how did we get here and what is left** (§8–12). It is meant to be read by someone who knows biology but not machine learning, or the reverse. A glossary (§13) and a question-and-answer section for non-ML readers (§14) are at the end.

Every factual statement carries one of three provenance levels:

tag	meaning
[verified]	checked directly by us — in the data files, in our own code and results, or in a primary source we opened
[chat]	stated in the planning conversation (the shared Claude chat) and <i>not</i> independently re-checked by us
[ours]	our reasoning, interpretation or design decision — arguable, and marked as such

Results carry one of three status levels: **final** (computed with the final code on the final data), **preliminary** (computed, but will be recomputed in the full three-representation audit that comes next), and **planned** (designed or implemented but not yet run).

A note on the planning chat. The chat (<https://claude.ai/share/057d4d81-f307-4f2b-aa1e-8eef3d6fea0b>) is a *frozen snapshot*: when I re-read it today it contained exactly the same 16 exchanges as before (ending with the PowerShell `curl` exchange), differing only in relative timestamps. The first message of the chat contained a pasted "brainstorming summary" that the share view hides ("Files hidden when shared"); what it contained can only be inferred from the replies (§9). If the chat was continued after it was shared, a new share link is needed for me to see the continuation.

1. The problem in one paragraph

Regulators and pharmaceutical companies are trying to replace animal toxicology with human-relevant laboratory methods called **new approach methodologies (NAMs)**. One of the most promising is **Cell Painting**: stain cells with fluorescent dyes, photograph them, and turn each image into thousands of numbers describing cell shape and texture. A machine-learning model can then read these numbers and predict whether a chemical is toxic or biologically active. The trouble is a well-documented failure mode: much of toxicity is, at bottom, *cells dying*, and **a model that merely counts how many cells survived** — no shape, no texture, no biology — can score surprisingly well on exactly these tasks. If a sophisticated morphology model does not beat that trivial counting model, it has shown nothing. Yet in practice nobody systematically checks, endpoint by endpoint, whether the morphology model really

beats the counting model, and checking properly is statistically delicate: there are hundreds of endpoints (so some "wins" occur by chance), many endpoints have almost no positive examples (so any number is noise), and the usual way of attaching a p-value to a win turns out to be wrong. **We are building a statistically rigorous, reusable audit that, for every assay endpoint, answers three questions: (1) does the full morphological profile beat a cell-count-only baseline after correcting for the number of endpoints tested; (2) is there enough data to even say; and (3) where morphology wins, is there a biologically coherent reason?**

2. The question, stated as simply as possible

When a Cell Painting model predicts a toxicity or bioactivity label, is it reading the cells' shape and texture — or only counting how many cells are left?

Everything else in this document is a way of answering that question honestly, endpoint by endpoint, with known error rates.

3. Background, in layers

3.1 Why toxicology is changing (the regulatory and scientific push)

- Animal testing is slow, expensive, ethically contested, and imperfectly predictive of human effects. Regulators and industry have therefore been encouraging **NAMs**: in-vitro assays, computational models and "organ-on-a-chip" systems that use human cells. **[verified]** The US FDA published a draft guidance on 19 March 2026 (Federal Register document 2026-05390), "*General Considerations for the Use of New Approach Methodologies in Drug Development*", which names four factors a sponsor should address when validating a NAM: **context of use, human biological relevance, technical characterization, and fit for purpose**. (We confirmed the title, date and the four factors from search results and the Federal Register listing; we could not open the page itself.)
- **[chat]** The chat adds that, as of January 2026, no organ-chip tool had been fully qualified through FDA's IStand pathway, and cites a US GAO finding that experts judged only 10–20 % of purchased human cells to be of sufficient quality for organ-chip studies. Neither figure was re-checked by us.
- The significance for us: *technical characterization* and *fit for purpose* are precisely the questions "is this assay's predictive power real, and for which purpose?" An audit that distinguishes real signal from a trivial confound is a direct instance of technical characterization.
- **Organ-on-a-chip (OoC)** is the competition's stated interest. Chips are microfluidic devices in which living human cells form tissue-like structures; they are imaged, and the images are analysed by AI. **[ours]** The same shortcut risk applies to any image-based chip readout, which is why we position the work as a *validation method* that chip assays should adopt — while stating plainly that our data are 2D plate cultures, not chips.

3.2 Image-based profiling and Cell Painting, step by step

Cell Painting is a standardised microscopy assay for taking a "morphological fingerprint" of cells.

1. **Culture and treat.** Cells are grown in multi-well plates (here 384-well) and exposed to a chemical at one of several concentrations for a set time (here 44 hours). **[verified]**

2. **Stain.** Fluorescent dyes label different cell compartments. The standard protocol uses six dyes imaged in five fluorescence channels covering nucleus (DNA), nucleoli/cytoplasmic RNA, endoplasmic reticulum, mitochondria, and the actin/Golgi/plasma-membrane group. **[verified]** In our data the CellProfiler feature names carry exactly these channel tags: DNA, RNA, ER, Mito, AGP (actin–Golgi–plasma membrane), plus a Brightfield channel.
3. **Image.** Each well is photographed at several sites. **[verified]** The image index lists about 2.0 million image files across 336,557 well–site records and six channels.
4. **Measure.** Software extracts features per cell — size, shape, intensity, texture, how the stains overlap, neighbour relationships — then aggregates them per well. **[verified]** For this dataset the authors computed profiles three ways: **CellProfiler** (classical, hand-engineered; 5,640 raw features), **CP-CNN** (a convolutional network trained on Cell Painting; 672 features) and **DINOv2** (a general-purpose vision transformer with no cell-specific training; 4,608 features).
5. **Normalise.** Every plate has DMSO (solvent) control wells; features are centred and scaled against them to remove plate-to-plate drift. **[verified]** (We re-implemented this from the authors' code.)
6. **Aggregate per compound.** The 8 concentrations × 2 replicates per compound are averaged (all wells, wells above the compound's point of departure, or wells in a window between two points of departure). The result is **one profile vector per compound**. **[verified]**

Why it is attractive: it is cheap, scalable, makes no assumption about a compound's mechanism, and its profiles are rich enough that compounds acting in similar ways look alike. That makes it useful for mechanism-of-action analysis, for detecting bioactivity earlier than cell death, and for toxicity prediction.

The key side-product: cell count. Because the software segments every cell, the number of cells per well comes for free. It is a one-number summary of how many cells survived.

3.3 The dataset: the Axiom OASIS hepatocyte screen (Ewald et al., *Cell Systems*, 2026)

[verified] Ewald, J.D. et al., *Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes*, *Cell Systems* 17(5), 101566 (2026), doi:10.1016/j.cels.2026.101566 (bioRxiv preprint January 2025; accepted 25 February 2026 per the chat **[chat]**).

aspect	what the study did
cells	primary human hepatocytes (liver cells taken from a donor, not an immortalised line); 2D cultures in 384-well plates
compounds	1,085 chemicals — pharmaceuticals, pesticides, industrial chemicals, food additives — chosen because in-vivo hepatotoxicity data exist for them [chat]
design	8 concentrations (about 0.01–100 µM), 2 replicates, 44 h exposure
readouts per well	LDH (an enzyme released when cell membranes rupture — a late-death marker); MT / RealTime-Glo (metabolic activity — an earlier stress marker); Cell Painting ; and cell count extracted from the images
outside labels	binary hit-calls for 402 ToxCast/Tox21 endpoints in three groups: 292 <i>cell-based</i> assays (specific biology in cell lines), 72 <i>cell-free</i> assays (purely biochemical: a protein or enzyme in a tube), 38 <i>cytotoxicity-burst</i> endpoints (the repository's file has 38 columns; the paper text refers to 48) [verified]
data on Zenodo	865 MB: profiles (three representations), metadata, an image index — all CC-BY 4.0. The raw images (≈650 TB [chat]) live in the public Cell Painting Gallery and are not needed. [verified]
code and labels	the authors' GitHub repository (BSD-3-Clause) contains the ToxCast binary labels, endpoint annotations (assay design, target family, cell line, tissue), and supplementary tables with the paper's hit calls and points of departure [verified]

What we actually model **[verified]**: 21,456 wells on 65 plates (4,849 DMSO controls), aggregated to **967 compounds** with a profile; after joining labels there are **405 endpoints** (292 + 72 + 38 + the paper's MT, LDH and cell-count hits), of which one (cell count) is used as a positive control.

3.4 What the paper claims — and which claim we are examining

As summarised in the chat from reading the paper **[chat]** (the underlying counts are consistent with the supplementary hit table **[verified]**: 198 cell-count hits, 377 MT, 130 LDH and 572 Cell Painting hits among 1,086 table rows):

1. Cytotoxicity-style readouts disagree: MT flags the most compounds (~40 %), cell count next (~20 %), LDH the fewest (~13 %).
2. **Cell Painting detects effects at far lower concentrations** than any of these: on average about **2.5× lower than MT, 8× lower than cell count, 16× lower than LDH**. (Sensitivity is measured by the *point of departure*, POD: the concentration at which the dose–response curve exceeds the 95th percentile of the controls. **[verified]** from the supplementary table's readme.)
3. **Morphology adds information beyond cell count — sometimes**. It beats a cell-count-plus-technical-features baseline for predicting MT, but not for predicting LDH.
4. The three feature-extraction methods perform about the same.
5. Against the 412 outside endpoints, Cell Painting predicts cytotoxicity and cell-based endpoints, but not cell-free ones — sensible, since it is a whole-cell assay — and **cell count alone also beats random for cytotoxicity endpoints**.

The claim our project examines is the second half of (3) and (5): how far is "morphology predicts toxicity" really "cell count predicts toxicity"? The paper makes this comparison, but (to our reading of its published code and metrics) it stops at reporting per-endpoint AUROCs; §3.6 lists what we think is missing.

3.5 Why the cell-count shortcut exists (mechanisms)

[ours], established general reasoning; not a claim about any specific compound:

- **Toxicity often is cell loss**. If a chemical kills or detaches cells, both the assay label ("toxic") and the cell count (fewer cells) change. A model needs no morphology to see this.
- **Many reporter assays are confounded by cytotoxicity**. In signal-decrease assays (for example "antagonist-mode" assays, where an active compound reduces a reporter signal) a cytotoxic compound lowers the signal for reasons unrelated to the target, so cytotoxic compounds look active.
- **Cell density affects almost everything**. Neighbour contacts, cell size and expression levels change with confluence; image features inherit that. Some of a "morphology" model's apparent skill may be cell density seen through another lens.
- **Technical effects** (plate, well position, batch) can correlate with labels if compound plating is not randomised.

[chat] The chat attributes the observation "many assay-activity benchmarks relate to cytotoxicity, so simply counting cells achieves unexpectedly high performance" to the JUMP consortium. What we could verify is a 2026 *Nature Communications* paper whose title states the claim: Seal, S. et al., "*Counting cells can accurately predict small-molecule bioactivity benchmarks*", Nat. Commun. 17, 2436 (2026), doi:10.1038/s41467-026-68725-5 **[verified via Crossref: title, authors, venue; we did not read the**

abstract], alongside the JUMP resource paper (Chandrasekaran et al., Nat. Methods 21, 1114–1121, 2024).

3.6 Where the existing analysis leaves gaps

[ours], from reading the authors' public code (`classifier/classify.py` , the compiled metrics) and the supplementary tables. These are our reading, and should be phrased cautiously in any write-up.

#	gap	why it matters
G1	No endpoint-level significance test against the baseline, and no correction for ~400 endpoints. Per-endpoint AUROCs are reported for the full model, the cell-count baseline and a random baseline.	With ~400 comparisons, many apparent "wins" occur by chance.
G2	Under-powered endpoints get a number like any other. In the data, 210 of 404 endpoints (52 %) have fewer than 15 active compounds among the compounds we model, and the smallest has 2.	An AUROC from 6 positives is mostly noise: we see AUROCs from 0.02 to 1.0 for such endpoints.
G3	The cell-count baseline is one number (mean cell count over the aggregated wells).	It throws away the dose–response shape: how steeply cells disappear, at which concentration. A strawman baseline makes morphology look better than it is.
G4	Cross-validation without shuffling. The authors use plain, unshuffled stratified k-fold on rows in a fixed order.	Folds are then contiguous blocks of compound IDs, a built-in distribution shift. We find it lowers the full model's AUROC by ≈ 0.04 relative to shuffled, repeated CV. It is a protocol effect, not an error.
G5	No look at <i>why</i> morphology wins beyond the target-family discussion of mispredictions.	The scientific value is in <i>where</i> morphology adds information.
G6	The headline sensitivity numbers (2.5x/8x/16x) rest on a "bioactivity POD" that is the minimum over many endpoints. For CellProfiler each compound can have up to 19 PODs (18 feature-category endpoints plus a global distance), and the lowest is used; LDH, MT and cell count each contribute a single POD.	A minimum over many correlated tests is biased low, so part of the "x more sensitive" claim may be selection. We implemented the audit for this (§6.8) but have not yet run it.
G7	Probability calibration is not reported.	A model can rank well and still be badly over-confident; any "confidence" shown to a user would be misleading.

4. The problem, formally

4.1 Objects

- **Compounds** $i = 1 \dots N$ ($N = 967$). Each has a **profile** $x_i \in \mathbb{R}^p$ (the per-compound aggregated morphological features; $p = 984$ for CellProfiler after filtering) and a **baseline vector** b_i containing *only cell-count information* (see §6.3).
- **Endpoints** $e = 1 \dots E$ ($E = 405$). Each gives a binary label $y_{ie} \in \{0, 1\}$ for the compounds on which the endpoint was measured (missing otherwise). n_{e^+} and n_{e^-} are the numbers of active and inactive compounds.

- A learner (XGBoost, hyper-parameters as in the paper) is trained on **profiles** to predict y_e (the *full* model f_{full}) and, separately, on **baseline vectors** (the *baseline* model f_{base}).
- The **performance measure** is the area under the ROC curve, AUROC (the probability that a random active compound is scored above a random inactive one; 0.5 = chance), estimated from **out-of-fold** predictions.

4.2 The estimand and the hypotheses

For endpoint e define the true effect $\Delta_e = \text{AUROC}_{full,e} - \text{AUROC}_{base,e}$ (the expected difference when both models are trained on comparable data and tested on new compounds).

- **H0 (no advantage):** $\Delta_e \leq 0$ — morphology gives no ranking benefit over cell count.
- **H1 (morphology advantage):** $\Delta_e > 0$.

We test H0 for every *powered* endpoint, control the **false discovery rate** across them (Benjamini–Hochberg at 5 %), and report every other endpoint as **indeterminate**.

4.3 The verdict taxonomy

verdict	condition	what it means
morphology_advantage	powered, $\hat{\Delta} > 0$, $q < 0.05$	we can say, with controlled error, that morphology beats cell count here
no_advantage	powered, otherwise	not shown; this bucket is split further below
indeterminate	fewer than 15 actives or 15 inactives	we refuse to claim anything either way
positive_control	the <code>cell_count</code> endpoint (its label is <i>defined</i> from the cell-count curve)	the strong baseline must reach $\text{AUROC} \approx 1$ here; it does (1.000) [verified, final]

The `no_advantage` bucket mixes two very different situations, so we also assign a **signal class** (descriptive, not a test; **preliminary**):

signal class	meaning
count_explained	the cell-count baseline is already informative (AUROC above the 95th percentile of what it achieves on permuted labels) and morphology is not shown to beat it — <i>the endpoint is cell counting in disguise</i>
morphology_signal_uncertified	the baseline is uninformative, morphology is informative, but it cannot be certified as an advantage
no_detectable_signal	neither model predicts the endpoint

This is the sharpest version of the chat's own phrasing: "*report which toxicity types are secretly just cell counting in disguise, and which genuinely need the full picture.*"

4.4 The six questions the project answers

#	question	answer type
Q1	For which endpoints does the full profile beat a cell-count-only model, after multiple-testing correction?	list of endpoints with q-values
Q2	For which endpoints can the question not be answered, because the data are too thin?	indeterminate list, with the reason
Q3	Where morphology wins, <i>why</i> : is it concentrated in a coherent assay or target family — or in particular kinds of compounds?	enrichment results, honest nulls included
Q4	Can the model's confidence be trusted?	calibration diagnostics per endpoint
Q5	Is the audit procedure itself valid — does it control its error rate, and recover truth when truth is known?	permutation controls, simulation with known ground truth
Q6	Is the result stable across feature representations, aggregation rules and evaluation protocols?	sensitivity and robustness tables

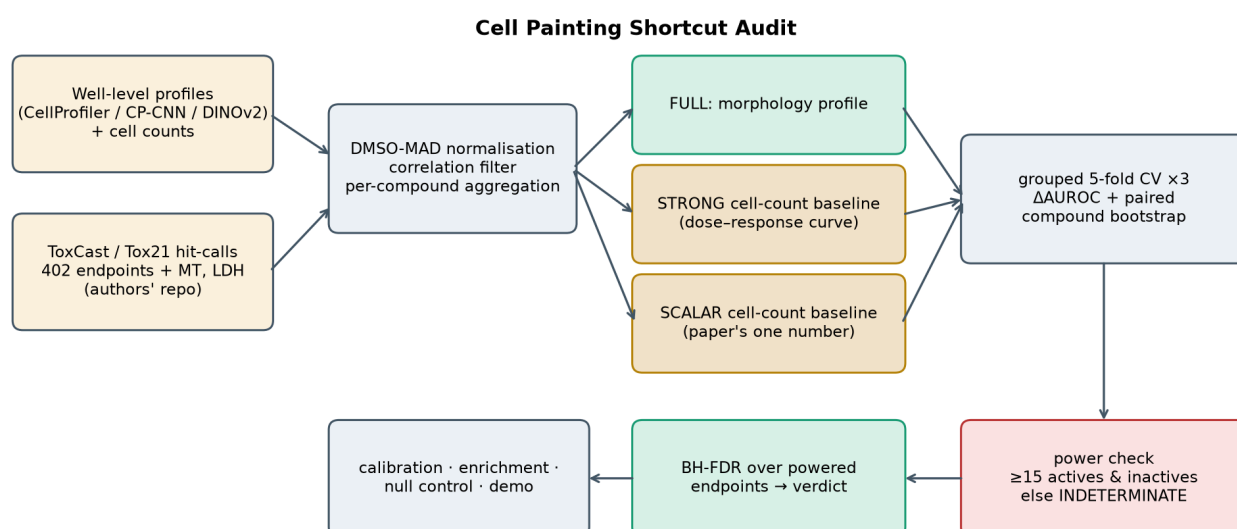
5. Why this is statistically hard (and what we do about each difficulty)

#	challenge	consequence if ignored	our response	evidence
C1	Hundreds of simultaneous tests	~5 % of null endpoints "win" by chance	Benjamini-Hochberg FDR across powered endpoints	final
C2	Tiny positive counts (median 15 actives for a cell-based endpoint; 8 for cell-free)	AUROC is noise; both false wins and false losses	explicit power rule (≥ 15 actives and ≥ 15 inactives) → <i>indeterminate</i> ; excluded from the FDR pool so they do not dilute power	final: 223 of 404 endpoints indeterminate
C3	Weak baseline = strawman	morphology looks better than it is	two baselines: the paper's scalar cell count and a <i>strong</i> cell-count-only model using the whole dose-response curve; the strong one is primary	final: 20 of 27 "wins" against the scalar baseline vanish against the strong one
C4	Replicate structure / leakage	duplicated compounds in train and test inflate scores	one row per compound; compound-grouped, stratified, shuffled CV repeated 3×, out-of-fold predictions only	final; unit-tested (duplicates with noise features score at chance)
C5	Analog series across folds (salts, isomers, drug classes)	chemical similarity inflates all models	Butina clusters of ECFP4 similarity as CV groups (921 clusters; 765 compounds sit in multi-member clusters; largest cluster 18, penicillins)	implemented; the grouped audit is planned
C6	The bootstrap p-value is anti-conservative (it resamples test compounds but ignores the variability of the fitted models)	too many certified endpoints	label-permutation null → empirical-null calibration (Efron-style): $p_{\text{cal}} = P(Z > z / \text{sd}_0)$, with sd_0 estimated from permuted-label runs	final: raw false-positive rate 13.2 % at nominal 5 %; calibrated 4.9 % on held-out runs
C7	Endpoints are correlated (shared compounds, related assays)	BH's guarantees assume independence or positive dependence	stated as a limitation; the permutation null is run through the <i>same</i> pipeline	limitation
C8	Class re-weighting makes probabilities over-confident	misleading "confidence" in the demo	calibration slope, ECE, Brier skill, cross-validated Platt recalibration	final: median slope 0.33
C9	Cell count is not the only possible shortcut (plate, well, batch)	a layout artefact could masquerade as morphology	separate technical-confound probe	final: plate/well/batch alone predict at chance (AUROC 0.510)
C10	Is the audit itself correct?	a method nobody validated	(a) permutation null; (b) positive control; (c) simulation with known ground truth (shortcut-only, morphology-adds, morphology-only, null endpoints)	(a)(b) final; (c) implemented, running
C11	The paper's headline sensitivity numbers may	a "× more sensitive" claim that is partly selection	paired POD fold audit: paper's definition vs single endpoints vs single-endpoint representations	implemented, planned

#	challenge	consequence if ignored	our response	evidence
	be biased by a minimum over many tests			
C12	"Morphology" is not orthogonal to density	an "advantage" may be a better density proxy	stated as a limitation; compound-level breakdown shows <i>where</i> the advantage lies (§7.5)	partly addressed

6. What we built to solve it

The code lives in `src/cpsa/`; numbered scripts in `scripts/` run each stage; everything is CPU-only, seeded, and covered by 44 unit tests.



6.1 Data layer

Loaders for the three profile representations, the ToxCast binary labels and annotations, and the paper's native hit calls; a re-implementation of the authors' normalisation and aggregation (per-plate DMSO median/MAD, correlation filter at $|r| > 0.9$, three aggregation rules). **Reproduction check [final]**: run under the paper's own protocol, our AUROCs differ from its published per-endpoint AUROCs by +0.004 (full model) and +0.007 (cell-count baseline) on average, Pearson $r = 0.79 / 0.70$ across endpoints (0.91 / 0.85 for endpoints with ≥ 30 actives).

6.2 The unit of analysis

One profile per compound, because the labels are compound-level (the same label would otherwise be copied onto 16 wells). This is the paper's own choice. The consequence **[ours]**: plate and well position are *not defined* for a compound averaged over 13–16 plates, so the plan's original "plate/well" baseline is replaced by a separate probe (§6.7).

6.3 The three models per endpoint

1. **Full** — the morphology profile (984 features for CellProfiler).
2. **Scalar cell-count baseline** — the paper's one number: mean cell count.
3. **Strong cell-count baseline** — *still only cell-count information*, but the whole dose-response: the cell count at each of 8 concentrations ($\div$ the plate's DMSO median), its minimum and mean, and the cell-count POD. This is the **primary comparator** because a verdict against a one-number baseline can be an artefact of that baseline being weak.

6.4 Evaluation and inference

Stratified, compound-grouped, shuffled 5-fold cross-validation repeated 3 $\times$; out-of-fold probabilities averaged over repeats; $\hat{\Delta}$ = pooled-AUROC difference; a paired bootstrap over compounds (1,000 resamples) for a standard error; $z = \hat{\Delta} / \text{SE}$; **calibration of z against the permutation null** (§5, C6); BH on the calibrated p-values over powered endpoints.

6.5 Verdicts and signal classes

As in §4.3. Raw (uncalibrated) verdicts are kept alongside, so the effect of the calibration is always visible.

6.6 Biology: where does morphology help?

- **Endpoint level:** one-sided Fisher exact tests, BH within each grouping, for target family, assay design, cell line, tissue and endpoint category, using EPA's own annotations rather than annotating 1,085 compounds by hand.
- **Compound level** (*preliminary*): a per-compound "morphology benefit" — how much better the full model ranks a compound than the cell-count baseline does — broken down by the compound's hit pattern across readouts and by vendor classes (pathway, target, research area, clinical phase).
- **Feature level:** gain importance aggregated by compartment (cells, nuclei, cytoplasm, image), feature type (granularity, texture, shape, ...) and channel. SHAP-based explanations for the certified endpoints are **planned**.

6.7 Reliability checks

Permutation null with split-half validation; the positive control; the technical probe; calibration diagnostics; sensitivity grids (power threshold, α , aggregation rule, chance-floored baseline, three representations); an exact reproduction of the paper's protocol.

6.8 New modules added after the first full run (this week)

module	purpose	status
signal_class	split "no advantage" into <i>count-explained</i> vs <i>no detectable signal</i>	implemented, run on two configurations (preliminary)
api (+ simulate)	dataset-agnostic entry point: any profile table + baseline table + binary labels → verdicts, with unit tests on synthetic data where the truth is known; a simulation study recovers power and false-credit rates	implemented and tested; the simulation is running
compound_classes	per-compound benefit and class breakdown (§7.5)	implemented, run on the primary configuration (preliminary)
chem_groups	chemical-similarity cluster CV groups (RDKit)	implemented; <code>--group-by chem_cluster</code> available; grouped audit planned
pod_audit	paired-fold audit of the 2.5×/8×/16× sensitivity claim	module and tests done; script and run planned
SHAP explanations, an FDA-structured "evidence pack" generator, optional conformal prediction sets	interpretability and regulatory packaging	planned

6.9 Deliverables built so far

A 17-page technical report (PDF), a Kaggle write-up draft, a demo-video script, a Streamlit app (endpoint explorer, compound explorer, audit table, enrichment and robustness, example images from the public gallery, method), a README, a pinned requirements file and a clean-environment dry run.

7. What we have found so far

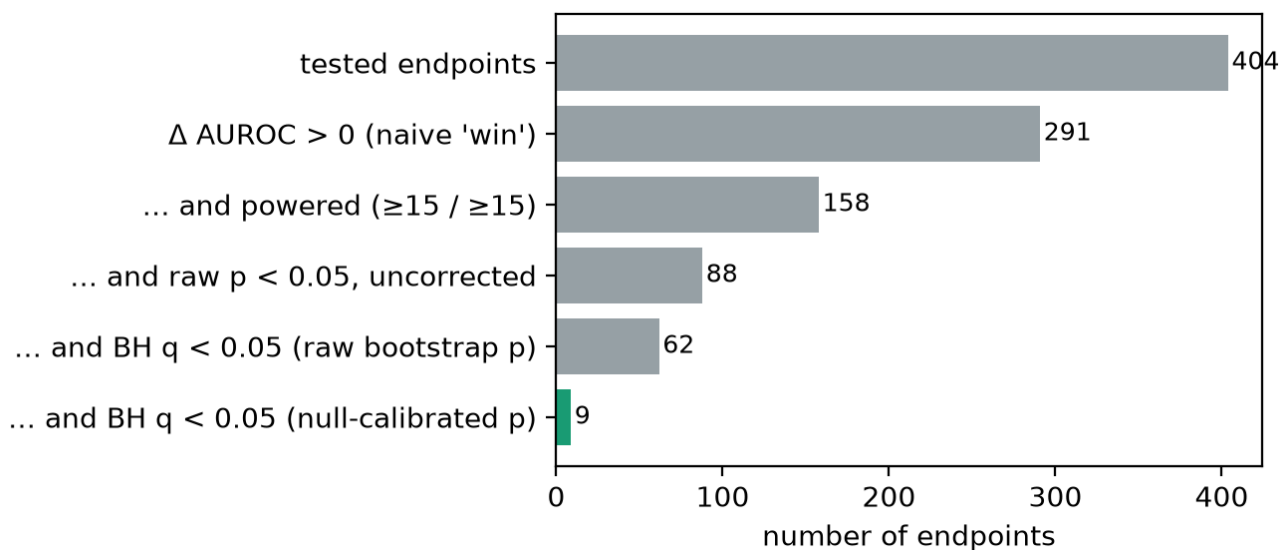
(Primary configuration: CellProfiler features, `allpod` aggregation. Final unless marked.)

7.1 Most endpoints cannot support a claim

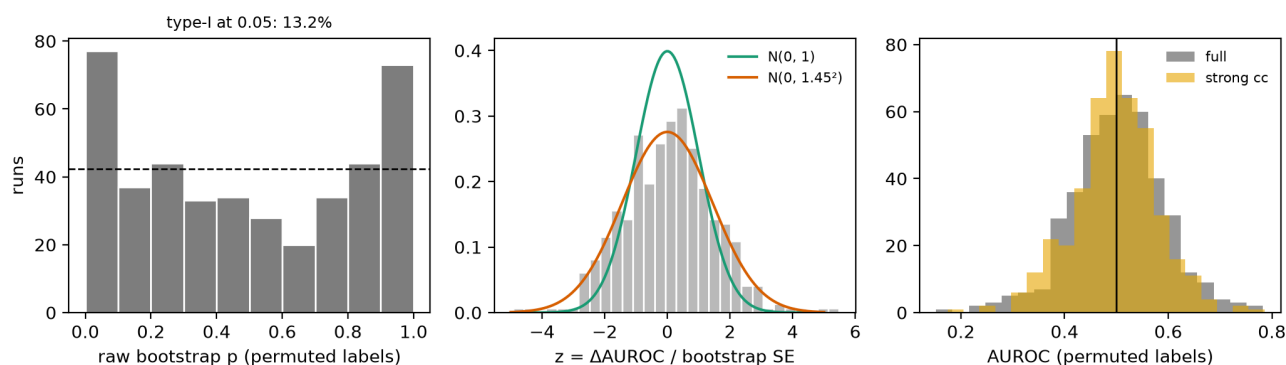
404 endpoints tested; **181 powered; 223 (55 %) indeterminate** (cell-based 145 of 292, cell-free 64 of 72, cytotoxicity 14 of 38). For indeterminate endpoints the full-model AUROC spreads far wider than for powered ones (sd 0.175 vs 0.117), and 56 of them lie outside 0.4–0.8.

7.2 The plain bootstrap would have over-called by a factor of about seven

step	endpoints
$\Delta\text{AUROC} > 0$ against the strong baseline ("naive win")	291
... and powered	158
... and raw $p < 0.05$	88
... and BH $q < 0.05$ on raw bootstrap p	62
... and BH $q < 0.05$ on null-calibrated p	9



The calibration is not cosmetic. In 424 powered label-permutation runs the z-score has sd 1.45 (95 % interval 1.34–1.56) where a valid test would give 1.0; the raw p-value rejects at nominal 5 % in **13.2 %** of runs. Calibrating with an sd estimated from the first 141 runs brings the false-positive rate on the other 283 runs to **4.9 %** (nominal 5 %). A matched-settings re-check (3 CV repeats, 1,000 resamples, 91 runs) gives sd 1.33 [1.13–1.51]; using 1.45 then yields 3.3 % — conservative. No permuted run produced a morphology_advantage verdict.



Median ΔAUROC over the 181 powered endpoints is +0.095 and 87 % are positive: the signal is broad, but only large effects can be certified one endpoint at a time at 15–60 actives.

7.3 A stable core of 9 across representations

representation	null sd ₀	certified vs strong baseline	vs scalar baseline
CellProfiler (primary)	1.45	9	27
CP-CNN	1.23	33	56
DINOv2 (null borrowed from CellProfiler; conservative)	1.45	40	39

All 9 CellProfiler-certified endpoints are certified under the other two representations too (union 46; 27 under at least two). The nine: PR-bla antagonist, GR-bla antagonist, PXR agonist, a BioMAP proliferation assay, and five cytotoxicity-burst endpoints (kidney, intestinal, HEK293, ME-180, ERR-HEK293T).

7.4 The paper's own readouts

readout	scalar cc	strong cc	full (CP)	Δ	certified?
MT	0.835	0.878	0.916	+0.037	CP-CNN and DINOv2 yes ($q \approx 0.005$); CellProfiler borderline ($q = 0.063$)
LDH	0.936	0.958	0.969	+0.011	never (q 0.14–0.30); but certified against the <i>scalar</i> baseline
cell_count (control)	0.975	1.000	0.991	—	control passes

This reproduces the paper's "morphology beats cell count for MT but not LDH" — and shows that the LDH conclusion *depends on baseline strength*: against the one-number baseline morphology looks better than it is.

7.5 Biology: a coherent family and a mechanistic signature

- **Endpoint level [final]:** cytotoxicity-burst endpoints are over-represented among advantage endpoints in all three representations (CellProfiler 5/24 vs 2.5 % elsewhere, $q = 0.007$; CP-CNN 10/24, $q = 0.010$; DINOv2 16/24, $q < 0.001$). Cell-free endpoints: **0 of the 8 testable** show an advantage. No other target family, cell line or tissue is enriched in any representation (liver-derived assays are *not* favoured).
- **Compound level [preliminary]:** among compounds that are *active* in an endpoint, the benefit of morphology over cell count is **+0.145 for compounds that react in Cell Painting but *not* in cell count, MT or LDH ("sub-lethal" compounds)**, +0.122 for MT/LDH-reactive compounds without cell loss, and only **+0.012 for compounds that cause cell loss**. In plain words: *where cells die, counting them already ranks the compounds; where cells stay alive but change, morphology adds the information*. This is the data's version of the explanation "morphology reads sub-lethal stress before cells are lost" — a mechanism-free test, because the compound classes come from readouts, not from assumptions about biology. It will be recomputed in the full audit.
- **Signal classes [preliminary]:** of 181 powered endpoints, **39 are count-explained, 51 show morphology signal that cannot be certified as an advantage, 82 show no detectable signal, 9 are certified advantage**. Most cytotoxicity-burst endpoints (16 of 24) are count-explained *and* are where morphology is additionally certified most often — consistent with the picture above: cell count solves most of the endpoint; morphology adds on top.

- **Feature level [final]:** importance is diffuse — nearly identical for advantage and non-advantage endpoints (granularity features carry 31 % of gain vs 25 % of features; image-level features 26 % vs 19 %): no single channel or compartment explains where morphology wins.

7.6 It is not a layout artefact

Plate, mean well row/column and batch alone predict the labels at chance (mean AUROC 0.510); adding them to the strong cell-count model changes nothing (0.564 vs 0.557); for none of the 9 certified endpoints does "technical + cell count" match the full profile.

7.7 Confidence is not trustworthy as delivered

For every powered endpoint the calibration slope is below 0.8 (median **0.33**, i.e. strongly over-confident); median expected calibration error 0.081; only 32 % of powered endpoints beat the prevalence predictor in Brier score. Cross-validated Platt recalibration lowers the median Brier score from 0.112 to 0.104; the demo shows recalibrated probabilities with each endpoint's calibration diagnostics.

7.8 Robustness [final]

Certified endpoints at minimum 10 / 15 / 20 / 30 positives per class: 8 / **9** / 16 / 24 (the threshold was fixed at 15 in advance and not tuned; a stricter threshold shrinks the BH pool and so raises the count). $\alpha = 0.10$: 30. Chance-floored baseline: 9 of 9 retained. Aggregation rule (vs strong baseline): all 18, allpod 9, allpodcc 0 — a stated limitation.

8. Scope, assumptions and threats to validity

What we claim: a method, and its results on one public dataset; that morphology's advantage over cell count is certifiable for a small, replicated core of endpoints; that plain bootstrap inference over-calls and a permutation-calibrated one does not; that endpoint-level and compound-level patterns point to sub-lethal stress as where morphology adds information.

What we explicitly do not claim: - Organ-on-a-chip results. The data are **2D hepatocyte plate cultures**; the work is positioned as a validation methodology that chip assays should adopt. - That the certified count (9 / 33 / 40) *estimates* how many endpoints benefit — it is a lower bound limited by power. - Anything about donor variability (single donor or pool; to be confirmed in the data audit), kinetics (one 44-hour time point), or in-vivo hepatotoxicity labels (the repository's `Invivo_ranking` column has no definition we could verify, so we did not use it).

Threats to validity (all stated in the report): 1. The empirical-null sd is an estimate (CellProfiler 424 runs, CP-CNN 190); DINOv2's null was not run. 2. BH under correlated endpoints. 3. "Morphology" and cell density are not orthogonal; the strong baseline sees only cell-count summaries. 4. The strong-baseline count depends on the aggregation rule. 5. Label heterogeneity (many assays, cell systems, ToxCast hit-calling). 6. Structurally related compounds across folds (addressed by the planned clustered-CV audit). 7. A baseline can always be made stronger; ours is a deliberate best-effort shortcut, not an upper bound.

9. How we got here (project history)

This section reconstructs the decision trail from the planning chat [\[chat\]](#) so that the reasoning is not lost.

9.1 The competition

AI for Life Science Challenge (5th Pazhou Algorithm Competition; Kaggle), **category Model & Algorithm**, deadline **10 October 2026**. The brief shows particular (not exclusive) interest in organ-on-a-chip, asks for solutions that are interpretable and applicable, and requires: a Kaggle write-up with a category declaration, a ≤ 5 -minute demo video, a public code repository, a technical report (15–20 pages), a registration form, and (optionally) a public demo. Scoring weights as recorded in the chat:

weight	criterion
30 %	Problem importance & impact
30 %	Technical approach & innovation
20 %	Results & validation
10 %	Reproducibility
10 %	Presentation

A team that includes a biologist earns a +0.5 bonus on the interpretability-and-reliability dimension (cross-disciplinary bonus).

9.2 The candidate ideas

The first message of the chat (hidden in the share) held the team's earlier brainstorming; the replies mention five ideas from it (ALS neurite phenotyping with leakage-aware evaluation, cardiotoxicity genetics, colon organoids, an image-quality-control tool for chips, and an endometriosis scorer). The chat then explored more; its final table covered fourteen candidates:

id	idea	data	outcome
A	ALS neurite phenotyping, leakage-aware re-evaluation	public (idr0112/Zenodo, 23.9 GB, CC-BY)	solid, bounded; small direct audience; risk that a leave-one-line-out result collapses to chance
J	Hepatocyte Cell Painting — cell-count shortcut audit	public (Zenodo 865 MB + Cell Painting Gallery)	chosen
D	Organ-chip image quality control (chip good/bad)	public Zenodo set (3,000+ brightfield images, expert labels)	a public repository had already tried it and reportedly no CNN met its bar; fallback only
F	Immune-cell tracking on a blood-brain-barrier chip (UFMTrack)	stated public by the authors	open-ended scope; strong incumbent tool
H	Flow-aware tracking	tiny dataset	method module at best
K, L, G	kidney chip; gut/lung barrier (TEER); vessel/barrier imaging	transcriptomic only / no open data found	real pain points, no usable data in the time available
M	Transfer of JUMP embeddings to chip brightfield images	no benchmark exists	too speculative for the time
N, O, P	multi-organ digital twin; FAIR-metadata tooling; chip-design simulation	none / off-skill	deprioritised
C, E	colon organoids; endometriosis scorer	unverified / none	not feasible

9.3 Why J

- **Impact (the researcher's-eye question).** J has the widest reach: Cell Painting is being adopted by industry consortia as a human-relevant, animal-sparing assay, and the cell-count-shortcut failure mode is general — so a method for detecting it is reusable. It also *corrects* something influential rather than adding yet another positive result.
- **Feasibility at nine days.** The decisive facts, resolved in the chat: the compiled profiles are only **865 MB**, pre-computed (no 650 TB image download, no weeks of feature extraction), and published under CC-BY.
- **The novelty trap, named early.** The paper already compares morphology with a cell-count baseline, so a reproduction is not a contribution. The plan therefore required three additions: multiple-testing correction, explicit "indeterminate" flagging for under-powered endpoints, and an enrichment analysis of where morphology wins. The chat's open risk — *where do the outcome labels live?* — was resolved during implementation: they ship in the authors' repository.
- **The domain-mismatch risk, named early.** 2D plates are not a chip. The chat advised arguing for relevance explicitly (a validation methodology organ-chip toxicology should adopt) rather than hoping judges would not notice. The report does this.

9.4 The plan and where we are against it

The chat's plan had a biology track (A1–A7: labels, tables, annotations, categories, enrichment, limitations, framing), an ML track (B1–B9: load, merge, baseline, full model, scale, FDR, power check, calibration, reusable script) and an integration track (C1–C5: demo, report, video, repo, dry run), with a go/no-go gate on day 5: *if almost nothing survives multiple-testing correction, that is still an honest*

finding but changes the story. That is almost exactly what happened: 62 raw "wins" became 9 after proper calibration.

item	status
A1–A7, B1–B9, C1–C2, C4–C5	done (C5 partly: clean install, tests, download and a pilot audit; the full audit has not yet been rerun from scratch)
C3 demo video	script written; recording is a team action
submission, repository publication, hosted demo, team declaration	team actions

10. Why it matters

Who is affected and what decision changes. - *Developers of image-based NAMs and phenotypic models* learn which endpoints a morphology model genuinely adds to, and which are cell counting in disguise. - *Regulators and sponsors* get a worked example of *technical characterization*: a structured, quantified statement of what a profiling assay can and cannot show. - *Organ-on-chip teams* get a reusable audit they can run on any dataset with profiles, a cell-count-like baseline and binary endpoints — the same shortcut risk applies to chips, where data are smaller and the temptation to over-claim larger. - *The Cell Painting community* gets a concrete demonstration of a statistical trap (an anti-conservative bootstrap p-value) and a cheap remedy.

Mapping to the FDA's four factors [ours]:

factor	what our work contributes
context of use	an explicit statement of what the audit applies to (endpoint-level claims in 2D primary hepatocytes), and which endpoints cannot be adjudicated
human biological relevance	primary human hepatocytes; no animal data; the biology analysis ties advantage to sub-lethal stress
technical characterization	reproduction of published numbers; positive control; permutation null with held-out validation; calibration; sensitivity analyses; technical-confound probe
fit for purpose	per-endpoint verdicts with uncertainty; explicit <i>indeterminate</i> ; calibration diagnostics instead of raw "confidence"

What makes it more than a re-analysis. The audit is validated on its own terms (simulation with known truth is in progress); its central statistical finding — raw bootstrap inference over-calls by a factor of 2–3 in error rate and ~7 in discoveries — is a transferable lesson; and the biology result is *replicated* across three independent feature representations.

11. Success criteria

Scientific. (1) The verdicts are reproducible and robust to representation and protocol; (2) the audit controls its false-discovery rate on null data and recovers truth on simulated data; (3) the biological explanation is supported by two independent angles (endpoint level and compound level); (4) every limitation that affects a claim is stated.

Competition (§9.1 weights).

criterion	our case	main risk
Importance (30 %)	regulatory timeliness (FDA draft guidance), generality of the shortcut problem	2D plates, not chips — mitigated by framing and explicit non-claims
Innovation (30 %)	a <i>validated</i> audit method; the permutation-calibration finding; strong-vs-weak baseline; compound-level mechanism test	the paper already made the basic comparison; innovation must be sold as the method, not the comparison
Results (20 %)	reproduction of the paper; replication across three representations; honest nulls	only 9 endpoints certified in the primary configuration may look thin — framed as a lower bound with a broad positive effect
Reproducibility (10 %)	CPU-only, seeded, MD5-checked, pinned, clean-environment run, 44 tests	the full audit takes ~30 minutes (CellProfiler) on 10 cores; documented
Presentation (10 %)	demo with endpoint/compound views and real images; video script ready	the video still has to be recorded

12. Open items and roadmap

Immediately (before the full audit). 1. Finish and read the simulation study (power and false-credit rates by effect size and number of actives). 2. Write the script for the POD sensitivity audit (the 2.5×/8×/16× claim) and run it from the supplementary tables. 3. Run the chemical-cluster grouped audit on the powered endpoints and compare with compound-grouped results (AUROC retention; does the core of 9 survive?). 4. SHAP explanations for the certified endpoints; the FDA-structured evidence pack; (optional) conformal prediction sets.

Then: the full three-representation audit. All 404 endpoints (not only powered ones) for CellProfiler, CP-CNN and DINOv2 with 3 CV repeats and 1,000 bootstrap resamples; a permutation null for **each** representation (including DINOv2, whose null was previously borrowed) at matched settings; all downstream tables (enrichment, compound classes, signal classes, sensitivity grids, figures, report numbers) regenerated from those outputs.

Finally. Update the report, Kaggle write-up and demo; record the video; publish the repository and the demo; fill in the team declaration; re-verify the FDA and Nature Communications citations against the primary pages; submit.

13. Glossary

- **AUROC** — probability that a random active compound is scored above a random inactive one. 0.5 is chance, 1.0 perfect.
- **Baseline (scalar / strong)** — a model allowed to see only cell-count information; scalar = one number; strong = the whole dose-response curve.
- **BH-FDR (Benjamini-Hochberg)** — a procedure that controls the expected fraction of false discoveries among the endpoints declared significant.
- **Calibration** — whether a stated probability of 70 % is right about 70 % of the time.
- **Cell-based vs cell-free endpoint** — a ToxCast assay run in living cells vs in a test tube with a purified protein or enzyme.

- **Count-explained** — an endpoint cell count already predicts about as well as morphology does.
 - **Cytotoxicity-burst endpoint** — ToxCast's measure of general cytotoxicity in a given cell type or tissue.
 - **DMSO** — the solvent used as the negative control on every plate.
 - **Empirical null** — the distribution of a test statistic when the true effect is zero, estimated by running the whole analysis on data with the labels shuffled.
 - **Endpoint** — one binary outcome label (a ToxCast assay hit-call, or the paper's MT/LDH/cell-count hit).
 - **Indeterminate** — too few positives or negatives to say anything.
 - **LDH / MT** — membrane-damage (late death) and metabolic-activity (earlier stress) assays.
 - **NAM** — new approach methodology: a non-animal method for safety assessment.
 - **Out-of-fold prediction** — a prediction made by a model that never saw that compound during training.
 - **POD (point of departure)** — the lowest concentration at which a readout departs from the controls.
 - **Powered** — at least 15 actives and 15 inactives.
 - **Profile** — the vector of image-derived features describing a compound's effect on cells.
 - **Shortcut** — a simple signal that predicts the label for reasons unrelated to the biology the model is supposed to capture.
-

14. Questions a biotech reader will probably ask

"Isn't it obvious that cell count predicts toxicity? Why is this interesting?" Because the question is *how much*, per endpoint, with error rates. Some endpoints are essentially cell-count readouts (our positive control is solved perfectly by counting); others are not. Knowing which is the point.

"Does morphology help or not?" Both, and that is the result: across powered endpoints morphology ranks compounds better in 87 % of cases, but the effect is individually certifiable only for a core of large effects, concentrated in cytotoxicity-burst endpoints; and at the compound level the benefit sits with compounds that alter cells *without* killing them.

"Why do half the endpoints get no verdict?" A ToxCast endpoint tested on a few hundred compounds with only 6–14 positives cannot support an AUROC comparison. Reporting a number anyway would be misleading; we saw AUROCs from 0.02 to 1.0 for such endpoints.

"Why is the bootstrap wrong?" It treats the trained models as fixed and only resamples the test compounds. In reality, a different training set gives a different model. With few positives this extra variability is large, so the bootstrap's confidence statements are too narrow. We measured this by shuffling labels and measured how much too narrow ($\approx 1.45\times$).

"Are the 9 endpoints biologically sensible?" Five are cytotoxicity-burst endpoints (the general-stress family where we expect and see an advantage). The four reporter assays — PR-bla antagonist, GR-bla antagonist, PXR agonist, and a BioMAP proliferation assay — are the ones we most want a biologist's judgment on: antagonist-mode assays can be cytotoxicity-confounded, and PXR is a xenobiotic-sensing receptor that hepatocytes express. We do not interpret them beyond reporting.

"Why not use the in-vivo liver-injury labels?" The annotation file has an `Invivo_ranking` column with values 1–4, but its definition is not documented in the repository, and the DILI identifiers carry no labels.

Guessing would be worse than omitting it.

"Does this transfer to organ-on-a-chip?" The *method* does, by construction: it needs profiles, a cell-count-like baseline and binary endpoints. The *results* do not: they are 2D hepatocytes.

"What would convince you this is wrong?" A simulation in which the audit credits morphology on shortcut-only endpoints; a chemical-cluster grouped audit in which the core of 9 collapses; or a POD audit showing the headline sensitivity numbers vanish for a single endpoint. The first two are implemented and about to be run; the third is next.

15. Sources and their verification status

source	what we used it for	status
Ewald, J.D. et al. <i>Cell Syst.</i> 17(5), 101566 (2026), doi:10.1016/j.cels.2026.101566	the study, data, claims	authors and venue verified (Crossref); paper summary [chat] ; underlying hit counts verified in the supplementary table
Zenodo 10.5281/zenodo.17067683 (CC-BY 4.0)	profiles, metadata, image index	verified (MD5-checked downloads)
github.com/jessica-ewald/2024_09_09_Axiom_OASIS (BSD-3-Clause)	ToxCast labels, annotations, supplementary tables, analysis code	verified
Seal, S. et al., <i>Nat. Commun.</i> 17, 2436 (2026), doi:10.1038/s41467-026-68725-5	cell count predicts bioactivity benchmarks	title, authors, venue verified (Crossref); content not read
Chandrasekaran, S.N. et al., <i>Nat. Methods</i> 21, 1114–1121 (2024), doi:10.1038/s41592-024-02241-6	the JUMP Cell Painting resource	verified (Crossref)
US FDA, <i>General Considerations for the Use of New Approach Methodologies in Drug Development</i> , Federal Register 2026-05390, 19 March 2026	four validation factors	title, date, factors verified from search results; page not opened
GAO statement on cell quality (10–20 %); ISTDAND status	context	[chat] only
Benjamini & Hochberg (1995); Efron (2004); Chen & Guestrin (2016)	statistical method; empirical-null idea; XGBoost	standard references
Cell Painting Gallery (cpg0037-oasis), AWS Open Data, CC0	a dozen example images for the demo	verified ; ≈650 TB figure [chat]

Appendix: where the numbers in this document live

quantity	file
all per-endpoint results (primary)	results/audit_table.csv
every headline number	results/cellprofiler_allpod/headline_numbers.json
null calibration	results/cellprofiler_allpod/null_calibration.json, null_control.csv
robustness	results/sensitivity_summary.csv, results/cellprofiler_allpod/sensitivity_postprocess.csv
enrichment	results/enrichment.csv and per-configuration copies
compound-level analysis (preliminary)	results/cellprofiler_allpod/compound_benefit.csv, compound_classes_*.csv
figures	results/figures/
the full technical report	report/technical_report.pdf